

DHIREN MAHAJAN

Software Developer with Experience in AI/ML

☎ 513-302-4035 - ✉ mahajadn@mail.uc.edu - 🔗 linkedin.com/in/dhirenmahajan - 🌐 github.com/dhirenmahajan - 🌐 dhirenmahajan.com

Experience

AI/ML Developer | Phillips Edison & Company

Jan 2025 – Aug 2025

- Architected Azure AI pipelines (Data Factory, Functions) ingesting 400+ daily emails/documents with 94% accuracy, eliminating 25 hours weekly manual sorting and improving SLA compliance from 72% to 96%
- Unified property data into curated datasets and SLA dashboards, reducing operational errors 60% and ticket resolution time 45% while delivering real-time visibility across 350+ properties worth \$4.68B
- Shipped RAG chatbot (GPT-4, LangChain, Azure) with 89% resolution, reducing response time from hours to seconds for 350+ properties
- Built PySpark jobs and Flask microservices that cut turnaround from hours to minutes, removing 95% manual touchpoints
- Added eval harnesses, safety guardrails, and feedback loops to enable compliant 24/7 self-service and improve answer quality

Software Engineering Co-op | Phillips Edison & Company

Aug 2023 – Aug 2024

- Delivered \$100K+ annual savings building Azure OpenAI pipelines reducing lease processing latency 90% (minutes to seconds per doc), processing 850+ documents monthly
- Automated approvals and cross-team workflows, cutting turnaround time 30% for new deals and critical changes
- Integrated SQL Server, Power Automate, and MRI ERP to automate property-management workflows and data movement
- Improved data accuracy 65% and eliminated 300+ manual hours per quarter through standardized validations and notifications
- Built full-stack platform (MRI, Python, SQL, Azure) serving 200+ users across 8 departments with 2,400+ monthly transactions

Firmware Engineering Co-op | Nokia

Jan 2023 – May 2023

- Implemented embedded C/C++ firmware for 800G fiber-optic modules across 15+ carrier networks, meeting sub-microsecond timing constraints and reducing defect rate from 12% to 3%
- Cut hardware bring-up time 40% with C#/.NET Core diagnostics dashboard for real-time telemetry and automated regression testing, accelerating \$50K+ quarterly production

Undergraduate Student Researcher | University of Cincinnati

Jan 2026 – Present

- Architecting Swift-based iOS app with CoreData and HealthKit for 500+ caregivers and autistic/IDD individuals, implementing rule-based emotional regulation tracking across 12+ self-regulation metrics with personalized coping strategy recommendations
- Designing HIPAA-compliant data architecture and privacy-first UI/UX projected to reduce caregiver burnout 35% and improve patient self-advocacy 28% based on pilot study benchmarks 145

Projects

DataGrep Pipeline Generator | FastAPI, Python, SQL, Docker, Supabase/Postgres, Slack

- Engineered AI pipeline generator converting natural-language to production Python/SQL workflows, processing concurrent queries and eliminating 2-3 day analytics bottleneck, reducing analyst workload 70% with 85% self-service adoption
- Built multi-source integration (4+ databases) with sandboxed Docker execution, OpenAI visualization, and Slack delivery serving multiple concurrent users at 99.7% uptime

GamesKart Data Platform | Supabase, Next.js, Postgres, Python

- Developed a commerce control plane modeling complex relationships between inventory, warehouses, and orders using PostgreSQL; wrote Python scripts to seed realistic fixture data for stress testing.
- Implemented row-level security policies and direct SQL API routes to enforce transactional integrity during checkout flows, ensuring zero data anomalies during concurrent write operations.

MAKEUC Hackathon Platform | TypeScript, GraphQL, Prisma, Postgres

- Led backend for 500+ participants across 120+ teams with real-time judging and leaderboard updates, processing \$25K prize distribution as production system for annual 24-hour event
- Optimized GraphQL resolvers with DataLoader batching and Redis caching achieving 67ms avg response time, handling 100+ concurrent users at 1,800+ requests/min during peak judging

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, C#, Java, C++, HTML, CSS

Frameworks: FastAPI, Django, Flask, Node.js, Express.js, React, Next.js, .NET Core, Spring Boot

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Redis

Cloud: AWS (Lambda, EC2, S3, RDS), Azure (Functions, SQL Database, Data Factory, Blob Storage), GCP

AI/ML: LangChain, OpenAI API, PyTorch, TensorFlow, PySpark, Pandas, NumPy, Scikit-learn

DevOps: Docker, Kubernetes, Git, GitHub Actions, Jenkins, CI/CD, Terraform, Linux

Education

University of Cincinnati

Aug 2021 – May 2026

Bachelor of Science in Computer Science & Certificate in Software Engineering

GPA: 3.994/4.0

- University Honors Scholar, International Outreach Scholar, UC Global Scholar; Dean's List each semester; 3 x Mantei Mae Award recipient for highest GPA in major; CEAS Ambassador; IEEE And ACM Executive Board Member